



LESSON - 14

SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

STRING FUNCTIONS

FUNCTION	DESCRIPTION	SYNTAX	EXAMPLE	OUTPUT
CONCAT	Concatenates two or more strings.	CONCAT(string1, string2, ...)	SELECT CONCAT('Hello, ', 'World!') AS greeting;	Hello, World!
SUBSTRING	Extracts a portion of a string.	SUBSTRING(string, start, length)	SELECT SUBSTRING('Hello, World!', 8, 5) AS extracted;	World
LENGTH	Returns the number of characters in a string.	LENGTH(string)	SELECT LENGTH('Hello') AS length;	5
UPPER	Converts a string to uppercase.	UPPER(string)	SELECT UPPER('hello') AS upper_text;	HELLO
LOWER	Converts a string to lowercase.	LOWER(string)	SELECT LOWER('WORLD') AS lower_text;	world
TRIM	Removes spaces or specified characters.	TRIM([chars FROM] string)	SELECT TRIM(' Hello World ') AS trimmed;	Hello World
REPLACE	Replaces occurrences of a substring.	REPLACE(string, search, replace)	SELECT REPLACE('Hello World', 'World', 'SQL') AS replaced;	Hello SQL
INSTR	Finds the position of a substring.	INSTR(string, substring)	SELECT INSTR('Hello World', 'World') AS position;	7